## Substitution method

**1**

**a** $x = -8, y = 8$

**b** $x = -5, y = 3$

**c** $x = -8, y = -3$

**d** $x = -6, y = -3$

**e** $x = -3, y = 5$

**f** $x = 6, y = 4$

**g** $x = 9, y = 5$

**h** $x = -2, y = 9$

**i** $x = 1, y = 6$

**j** $x = 8, y = 4$

**k** $x = -5, y = -8$

**l** $x = 8, y = 1$

**m** $x = -\frac{4}{15}, y = \frac{13}{3}$

**n** $x = 7, y = 4$

**2**

**a** $x = \frac{4 + 7y}{3}$

**b** Infinitely many solutions.

**c** Dependent

**3**

**a** $y = -\frac{64}{3} - \frac{5x}{3}$

**b** $x = -8$

**c** $y = -8$

**4**

**a** $x = 7, y = 4$

**b** $x = -6, y = 5$

**c** $x = 6, y = -5$

**d** $x = 2, y = 3$

**e** $f = \frac{2}{5}, g = \frac{6}{5}$

**f** $q = \frac{4}{5}, p = \frac{1}{8}$

**g** $a = \frac{3}{2}, b = \frac{5}{2}$

**h** $v = \frac{2}{5}, t = \frac{5}{4}$

**5**

**a** $x = 4, y = 2$

**b** Yes

**c** Yes

## Applications

**6**

**a** $x = y + 10$

**b** $x + y = 28$

**c** $y = 9$

**d** $x = 19$

**7**

**a** $y = x + 6$

**b** $x + y = 22$

**c** $x = 8$

**d** $y = 14$

**8**

**a** $y = 10x, y = 7x + 12$

**b** $x = 4$

**c** $y = 40$

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**9**

**a** $y = 5x, y = 9x - 32$ **b** $x = 8$

**c** $y = 40$

**10**

**a** $x + y = 3000, 0.06x + 0.08y = 204$ **b** $x = \$1800$ and $y = \$1200$

**11**

**a** $2f + 3g = 10.08$ and $5f + 7g = \$24.08$ **b** $f = \$1.68$ and $g = \$2.24$

**12**

**a** $y = 4x - 190\,000,\ y = x + 560\,000$ **b** $x = 250\,000$ and $y = 810\,000$

**13**

**a** $x + 10y = 105$ and $x + 15y = 155$ **b** $x = 5,\ y = 10$

**c** $\$255$

**14** Each racket is $\$500$ and each ball is $\$50$.

**15** $15$ pieces of $10c$ coins and $45$ pieces of $5c$ coins

**16** $35$ hotdogs and $52$ sodas.

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